My degree from DTH – civilingeniør in acoustics, 1989, revisited and brought up to date

In January 1989 I was awarded my diploma as *civilingeniør* – Master of Science in Engineering – by the Technical University of Denmark (DTH, today DTU) in Lyngby, with a thesis at the Acoustics Laboratory. As with the courses of my B.Sc. from Sussex, I have recreated the courses – not as exam notes, but as they look today: with the physics and history behind them, interactive figures to experiment with, and what has happened in each field since 1989. Every page can also be downloaded as a PDF.

DTH in the 1980s

The Technical University of Denmark was founded in 1829 as the Polytechnic Institute (*Den Polytekniske Læreanstalt*), with H. C. Ørsted as its first director. In the 1960s and 70s it moved from Copenhagen to the large campus at Lyngby, and in 1994 it took the name Danmarks Tekniske Universitet (DTU).

In the 1980s there was no bachelor's degree in Denmark, and the engineering degree was not divided into a bachelor's and a master's stage. It was a single, continuous course of study, normally lasting five years and requiring 330 points, which led to the title *civilingeniør, polyteknisk kandidat* – rendered on the diploma in English as Master of Science in Engineering. The course was built up of courses that each carried a number of points – most of them 6 – and students could put together much of their programme themselves within a field of study. A thesis completed the degree. The split into bachelor's and master's degrees came only later.

The Acoustics Laboratory was a strong research environment with close ties to Brüel & Kjær, Bang & Olufsen and the hearing-aid industry. In the early 1980s it helped develop the measurement of sound intensity with two microphones, and in the middle of the decade work began on the room-acoustics program ODEON. Today acoustics at DTU is part of DTU Electro.

The special study plan

I came to DTH with a B.Sc. from the University of Sussex. As there was no Danish bachelor's stage to be admitted to, I applied for a dispensation, and on 4 December 1984 the Polytechnic Institute approved a *special study plan* (*særstudieplan*) for me. I was not, however, given credit for all my Sussex courses – partly because it was a science degree rather than an engineering degree. The special study plan required at least 198 points, of which at least 30 for the thesis, and the basic courses that the Sussex degree covered – mathematical analysis and linear algebra, mechanics, thermodynamics, electromagnetism, quantum mechanics, atomic and nuclear physics, solid-state physics, relativity, wave physics, laser engineering and plasma physics 1 – carried no points for me. For comparison, the engineering degree normally required 330 points. The diploma states that the

degree was completed under the special study plan with a reduced requirement of 168 points. Grades were given on the Danish 13-point scale. The diploma of 31 January 1989 declares me *civilingeniør, polyteknisk kandidat* – in English, Master of Science in Engineering.

> My supervisor for the thesis, from September 1988 to January 1989, was **Finn Jacobsen**, one of the pioneers of sound-intensity measurement, whose lecture notes on sound propagation in pipes and ducts are among the thesis's sources.

The courses are arranged here by subject rather than by year. Fifteen of the pages were written for this collection; three courses – plasma physics, electron spin resonance and probability – already had their own pages on the site.

Acoustics

Fundamentals of acoustics

The wave equation, sound pressure and decibels, impedance and intensity, the point source, A-weighting, standing waves in tubes and absorption in air.

8042 · 1987 · 6 points

PDF

Since 1989: FEM/BEM simulation, MEMS microphones, acoustic cameras and active noise cancellation in headphones.

Machinery noise in industry

Noise sources in machines, sound-power measurement, direct and diffuse field in a factory hall, enclosures, vibration isolation and noise limits.

8044 · 1988 · 6 points

PDF

Since 1989: The EU noise directive (2003), variable-speed drives and condition monitoring by vibration analysis.

Traffic noise and housing

Tyre and engine noise, the line source, L_{den}, noise barriers and Maekawa's formula, façade insulation and health effects.

5148 · 1988 · 3 points

PDF

Since 1989: Noise maps, CNOSSOS-EU, low-noise asphalt – and evidence of traffic noise's effect on the heart and on sleep.

Building and room acoustics

Sabine and reverberation time, absorbers, room modes, concert halls, the mass law and coincidence, double walls and impact sound. Also covers the project course in building acoustics.

5142 · 1984 · 6 points · 5147 · 1988 · 6 points

PDF

Since 1989: Auralisation with ODEON, sound classes in DS 490 and acoustics in tall timber buildings.

Acoustic communication

The ear and the basilar membrane, loudness and masking, the source–filter model of speech and formants, STI and hearing loss.

5131 · 1987 · 6 points

PDF

Since 1989: Digital hearing aids (1996), cochlear implants and MP3 built on psychoacoustics.

Electroacoustics and advanced electroacoustics

The analogies, transducer principles, the loudspeaker, Thiele–Small parameters, closed and vented boxes, microphones and calibration.

5121 · 1987 · 6 points · 5122 · 1987 · 6 points

PDF

Since 1989: DSP-controlled active loudspeakers, MEMS microphones and class-D amplifiers.

Signal analysis

The Fourier transform, sampling and aliasing, the DFT and FFT, windows and leakage, correlation, frequency responses and coherence.

5161 · 1988 · 6 points

PDF

Since 1989: 24-bit sigma–delta converters, swept-sine measurements and compressed sensing.

Ultrasound and materials

Ultrasound

Piezoelectric transducers, reflection at interfaces, near field and far field, attenuation, pulse–echo, Doppler and high-power ultrasound.

1846 · 1985 · 6 points

PDF

Since 1989: 3D scanning, pocket scanners, elastography and ultrafast imaging – with contributions from DTU.

Non-destructive testing

Fracture mechanics and fatigue, dye penetrant and magnetic particles, eddy currents, the ultrasonic A-scan, radiography and acoustic emission.

1845 · 1987 · 6 points

PDF

Since 1989: Phased arrays, TOFD, industrial CT and permanent structural health monitoring.

Electron microscopy and microanalysis

The electron wavelength, magnetic lenses, TEM and SEM, the interaction volume, EDS and WDS, and Moseley’s law.

1848 · 1987 · 6 points

PDF

Since 1989: Aberration correction (1998) and cryo-electron microscopy (Nobel Prize 2017).

Electrical engineering

Circuit theory

Kirchhoff, Thévenin and Norton, transients, phasors and impedances, resonance, filters and Bode plots, the operational amplifier. Also covers the laboratory course in circuit theory and electronics.

4901 · 1986 · 6 points · 4932 · 1985 · 3 points

PDF

Since 1989: Free SPICE simulation, switch-mode power supplies and SMD components.

Digital electronics

Boole and Shannon, binary numbers, logic gates, Karnaugh maps, adders, flip-flops and counters, TTL and CMOS, A/D and D/A conversion.

4303 · 1985 · 6 points

PDF

Since 1989: Moore's law from a million to over 100 billion transistors, FPGAs and microcontrollers.

Applied field theory

Maxwell's equations, Laplace and boundary-value problems, relaxation, skin effect, transmission lines and the Smith chart, waveguides and antennas.

4820 · 1987 · 6 points

PDF

Since 1989: Computational electromagnetics, signal integrity in GHz electronics and metamaterials.

Physics, mathematics and chemistry

Plasma physics 2

Debye shielding, the plasma frequency, particle drifts, magnetohydrodynamics, fusion, the tokamak and JET.

4023 · 1985 · 6 points

PDF · also part of the Sussex courses

Since 1989: JET's energy record in 2023, ITER under construction and private fusion companies.

Electron spin resonance

An interactive refresher on ESR (in Danish): Zeeman splitting, the resonance condition, the Bloch equations, hyperfine structure and pulsed ESR – from Zavoisky in 1944 to today.

4087 · 1985 · 6 points

Since 1989: Pulsed ESR and ESR on single spins in diamond (NV centres) for quantum sensors.

Probability

Calculating with uncertainty: sample spaces and Kolmogorov's axioms, conditional probability, combinatorics, expectation and common fallacies.

0142 · 1986 · 6 points

PDF · Distributions · Bayes

Since 1989: Bayesian statistics and Monte Carlo methods have become everyday tools thanks to cheap computing power.

Basic chemistry

Stoichiometry, thermochemistry, equilibrium, acids and bases, titration, electrochemistry and corrosion, reaction kinetics.

2010 · 1986 · 6 points

PDF · Le Chatelier

Since 1989: The lithium-ion battery, Power-to-X and green chemistry.

Projects and laboratory courses

Besides the theoretical courses the degree included a number of project and laboratory courses that have not been given pages of their own, but whose material appears on the pages above:

- **Projects in acoustics** (5191, 1984, 6 points) – a project course at the Acoustics Laboratory. See fundamentals of acoustics and room acoustics.
- **Building-acoustics design** (5147, 1988, 6 points, integrated course) – treated together with building and room acoustics.
- **Laboratory exercises in electrophysics** (4077, 1986, 6 points) – see circuit theory and field theory.
- **Laboratory exercises in circuit theory and electronics** (4932, 1985, 3 points) – treated together with circuit theory.
- **Individual course at the Department of Process Engineering** (1988, 6 points).

The thesis

Sound propagation in ducts with absorbent linings

Silencers and sound propagation in ducts with porous linings: the theories of Morse and Scott, Delany–Bazley, Newton–Raphson and Muller, the anomalous (0,0) and (0,2) modes, and measurements on a silencer. Supervisor: Finn Jacobsen.

Acoustics Laboratory · 1989 · 36 points

PDF (summary) · the thesis · Fig. 3.10(b) · the program “Extended”

Since 1989: FEM modelling of silencers, new material models (Miki, Johnson–Champoux–Allard) and acoustic metamaterials.

The courses on the diploma

The supplement to the diploma of 31 January 1989 lists 22 courses and the thesis, a total of 168 points under the special study plan. Course titles are translated; the Danish originals are in brackets.

No.	Points	Course	Page
0142/86	6	Probability (Sandsynlighedsregning)	Probability
1845/87	6	Non-destructive testing (Ikke-destruktiv prøvning)	Non-destructive testing
1846/85	6	Ultrasound (Ultralyd)	Ultrasound
1848/87	6	Electron microscopy and microanalysis (Elektronmikroskopi og mikroanalyse)	Electron microscopy
2010/86	6	Basic chemistry for civil, electrical and mechanical engineering (Kemisk grundkursus for B-, E- og M-linierne)	Basic chemistry
4023/85	6	Plasma physics 2 (Plasmafysik 2)	Plasma physics
4077/86	6	Laboratory exercises in electrophysics (Elektrofysiske øvelser)	–
4087/85	6	Electron spin resonance (Elektronspinresonans)	ESR
4303/85	6	Elementary digital electronics (Elementær digitalelektronik)	Digital electronics
4820/87	6	Applied field theory (Anvendt feltteori)	Applied field theory
4901/86	6	Elementary circuit theory (Elementær kredsløbsteori)	Circuit theory
4932/85	3	Exercises in circuit theory and electronics (Øvelser i kredsløbsteori og elektronik)	Circuit theory
5121/87	6	Electroacoustics (Elektroakustik)	Electroacoustics
5122/87	6	Advanced electroacoustics (Videregående elektroakustik)	Electroacoustics
5131/87	6	Acoustic communication (Akustisk kommunikation)	Acoustic communication
5142/84	6	Building and room acoustics (Bygnings- og rumakustik)	Building and room acoustics
5147/88	6	Building-acoustics design, integrated course (Bygningsakustisk projektering)	Building and room acoustics
5148/88	3	Traffic noise and housing (Trafikstøj og boliger)	Traffic noise
5161/88	6	Signal analysis (Signalanalyse)	Signal analysis
5191/84	6	Projects in acoustics (Projekter i akustik)	–
8042/87	6	Fundamentals of acoustics for mechanical engineers (Grundkursus i akustik for maskiningeniører)	Fundamentals of acoustics
8044/88	6	Machinery noise in industry (Maskinstøj i industrien)	Machinery noise
80—/88	6	Individual course, Department of Process Engineering (Individ. kursus ved Procesteknisk Institut)	–
–/89	36	Thesis, Acoustics Laboratory: Sound propagation in ducts with absorbent linings (Lydtransmission i absorptionsbeklædte rør)	The thesis

What has happened since 1989?

The biggest change is computing power. In 1989 I solved Scott's equation frequency by frequency in HP BASIC; today a laptop computes whole sound fields in three dimensions by the finite element method, simulates concert halls so that they can be heard before they are built, and analyses measurements in real time. The measurement microphone has been joined by MEMS microphones in billions of phones, analogue electronics by signal processors, and hearing aids, loudspeakers and ultrasound scanners have become small computers. The electron microscope can now see single atoms, and chemistry has taken a leading role in the green transition. But the foundations – the wave equation, Maxwell's equations, Fourier and Kirchhoff – are the same, and that is what these pages are about.